

Ankur Vadlamani

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Education

Ecole Centrale de Lille

Semester 8 exchange student

Lille, France

Feb. 2026 – Jun. 2026

- **Coursework:** Robotized Production Line, Electronics for Biomedical Engineering, Rapid Prototyping of Sensors

Mahindra University

B.Tech in Electronics and Computer Engineering

Hyderabad, India

Sept. 2022 – Aug. 2026

- **GPA:** 7.67/10
- **Coursework:** Programmable Devices, Analog & Digital Circuits, Digital System Design, VLSI Design, Optoelectronic Devices, Computer Architecture, Machine Learning, Data Structures & Algorithms

Publications & Conferences

Comprehensive Design and High-Performance Control of an AUV

Sri Sai Deep Duduka, Amit Saraf, Ankur Vadlamani, Sumanth Kandikattu, Gopinath G R

Mar. 2026

- Presented at RoboVis 2026 [[Link to the conference website](#)]
Topics: autonomous systems, embedded control, sensor fusion, system validation

Technical Skills

- **Programming:** C, Python, MATLAB, Verilog, Bash, LATEX
- **Embedded:** ESP32, RP2040, Arduino, Raspberry Pi, NVIDIA Jetson Xavier, UART, SPI, I2C, CAN
- **Electronics:** Analog/Digital Circuit Design, PCB Design, Sensor Interfacing, Signal Conditioning
- **Tools:** LTspice, Cadence Virtuoso, KiCAD, GNU Radio, Git/GitHub, Linux

Experience

QUBEATS

Engineering Intern

Hyderabad, India

Jun. 2025 – Aug. 2025

- Established a complete sensor testing laboratory from scratch, enabling systematic validation and data acquisition.
- Designed and fabricated two custom PCB development boards, enabling rapid multisensor integration and reducing prototype development time.
- Tested and calibrated navigation sensors including HMR2300, GPS, and barometer modules, improving measurement reliability for autonomous navigation experiments.
- Implemented Kalman filtering and repeatable experimental testing procedures to reduce sensor noise and improve measurement stability.

Defence research & Development Laboratory (DRDL), DRDO

Research Intern

Hyderabad, India

Jul. 2024 – Aug. 2024

- Tested PlutoSDR using GNU Radio, creating FM receiver and transmitter flowgraphs for validating SDR functionality.
- Migrated to PyADI based Python workflows, enabling automated testing and repeatable data acquisition pipelines.
- Developed and experimentally validated a custom PlutoSDR based transceiver module and implemented FM-based D2D communication protocol for UAV applications.

- Prepared comprehensive documentation and experiment reports, facilitating knowledge transfer within the team.

Research & Projects

Distributed State Estimation for Vehicle Fleets

Feb. 2026 – Jun. 2026

- Developed distributed observers for lateral vehicle dynamics using LMI-based observer synthesis.
- Implemented Luenberger observers in MATLAB for estimating vehicle slip angle and yaw rate.
- Validated observer performance using Renault Zoe experimental datasets, demonstrating robust vehicle state estimation under distributed communication topologies.

Electrochemical Impedance Spectroscopy (EIS) System

Jun. 2026

- Designed and experimentally validated an analog Electrochemical Impedance Spectroscopy (EIS) measurement system for estimating impedance over a 10 Hz–1 MHz frequency range.
- Developed the complete signal-conditioning chain using OP07, INA128, AD633 lock-in amplifiers, RC low-pass filters, and Arduino, with circuit verification in LTspice.
- Validated impedance magnitude and phase estimation by comparing theoretical, LTspice, and experimental measurements across the operating frequency range.

ESP32-based PLC design

Sept. 2025 – Dec. 2025

- Under guidance of Dr. Bhargava Rajaram, developed the architecture of an ESP-32 based programmable logic controller with isolated digital I/O over CAN, emulating industrial PLC standards.
- Integrated wireless communication capabilities, including Wi-Fi and BLE, to enable remote monitoring, configuration, and control.
- Designed and validated the PLC prototype, demonstrating reliable isolated digital I/O and CAN-based communication.

Autonomous Underwater Vehicle

Oct. 2023 – Mar. 2025

- Under guidance of Dr. Gopinath and Dr. Sebastian, designed, fabricated, and tested the electrical system of the AUV, which helped us reach the finals of SAUVC Singapore Challenge 2025.
- Designed a power distribution network with safety relays and kill-switch interface, and developed a custom Raspberry Pi Pico shield to interface sensors with the NVIDIA Jetson Xavier, ensuring reliable operation under mission constraints.
- Designed custom PCB interfaces for sensor-to-Jetson Xavier communication, improving system integration and deployment reliability.

Achievements

SAUVC 2025 Finalist | *Singapore polytechnic, Singapore*

Mar. 2025

- Ranked top 6 out of 45+ international teams by designing, fabricating, and testing a fully functional AUV.

LAM Research Challenge Finalist | *Indian Institute of Science, Bengaluru*

Nov. 2024

- Ranked top 25 out of 600+ nationwide teams by leading systems engineering, system integration, and rapid prototyping of a complex mechatronic project.

Merit Scholarship | *Mahindra University, Hyderabad*

2023

- Recognized for outstanding academic achievement with a merit scholarship after ranking in the top 10% of my class for the 2022-23 academic year.

Positions of Responsibility

AUV Club — Mahindra University

- **Electrical Head (Apr. 2024 – Apr. 2025):** Led a team of 15 juniors, managing the entire electrical system of the AUV, and organized workshops on circuit design, PCB design, and microcontrollers.
- **Mentor (2025 – Present):** Mentoring the next AUV team for competitions such as TAC and RoboSub, providing guidance on electrical design, sensor integration, and system-level prototyping.